Technology acceptance models in healthcare

"Studies evaluating clinical decision support or machine learning tools with health professionals using TAM, TAM2 or UTAUT. Return a matrix with: authors, year, journal, DOI, tool evaluated, participants, model used, constructs measured, and main findings. Limit to 2018-2026."
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The retrieved records support a focused matrix of **direct empirical studies** from 2018-2026 on **clinical decision support systems and machine-learning tools** evaluated with **health professionals** using **TAM, TAM2, UTAUT, or an adapted UTAUT/TAM variant**. The excerpts do **not expose author names**, so the table below uses the bibliographic fields that are visible in the provided material.

Authors	Year	Journal	DOI	Tool evaluated	Participants	Model used	Constructs measured	Main findings
Not shown in excerpt	2025	BMJ Health and Care Informatics	10.1136/bmjhci-2024-101128	CDSS alerts	72 physicians in outpatient departments of a single academic medical centre in northern Taiwan	TAM	PEOU, PU, attitude toward usage, user satisfaction, behavioural intention; plus external factors such as age, clinical experience, CDSS operating status, patient volume, consultation frequency, gender, specialty [1] [2]	PEOU increased PU, attitude, and user satisfaction; age and patient volume reduced PU/PEOU; clinical experience improved them; user satisfaction did **not** predict behavioural intention, while PEOU did [1] [2].
Not shown in excerpt	2026	JMIR Medical Informatics	10.2196/80527	ML-based CDSS for urinary tract infection care	263 general practitioners and assistants were surveyed after implementation; 34 practices used the software [3]	UTAUT-based survey	Trust, perceived usefulness, understandability of output, workflow fit/ease of integration, perceived improvement in care/outcomes [3]	Most respondents trusted and understood the tool; many found it useful and decision-supportive, but uncertainty remained about patient outcome improvement; only 48% felt it was easy to integrate into workflow [3].
Not shown in excerpt	2023	BMC Medical Informatics and Decision Making	10.1186/s12911-023-02101-x	NLP-driven / ML-based CDSS for concept-based searching in hospital practice	Clinicians in a Norwegian hospital; mixed-methods evaluation of adoption [4]	UTAUT	Acceptance/attitude, intention to use, and UTAUT domains used as the analytic lens [4]	Clinicians reported high acceptance and positive future-use intention; the main perceived benefit was improved allergy detection and patient safety, while implementation improvements were needed for alarms, integration, and point-of-care access [4].
Not shown in excerpt	2021	Journal of Medical Internet Research	10.2196/25929	AI-enabled CDSS evaluation framework	156 respondents completed the survey [5]	Acceptance framework informed by system quality, information quality, service quality, perceived ease of use, perceived benefit, and user acceptance [5]	User acceptance, perceived ease of use, information quality, service quality, perceived benefit, system quality [5]	User acceptance was the central success dimension; it was directly driven by perceived ease of use, information quality, service quality, and perceived benefit, with the final model explaining 89% of variance in user acceptance [5].
Not shown in excerpt	2026	Frontiers in Digital Health	10.3389/fdgth.2026.1896184	LLM-based CDSS for gynecologic oncology	29 physicians in a tertiary hospital clinic [6]	TAM plus UTAUT characteristics	Questionnaire categories derived from TAM/UTAUT; focus on willingness to use, transparency, evidence integration, validation, physician oversight, accountability [6]	No questionnaire category significantly predicted willingness in OLS; descriptive results showed a generally positive but cautious attitude, with acceptance depending on transparency, clinical validation, and preserving physician

								oversight [6].
Not shown in excerpt	2026	Health Policy and Technology	10.1016/j.hlpt.2026.101286	AI-CDSS across five clinical tasks	551 healthcare professionals from six hospitals in five Chinese cities [7]	UTAUT integrated with Threat Avoidance Theory and endowment effect theory	Performance expectancy, social influence, functional threats, endowment-based threats, resistance, behavioural intention [7]	Performance expectancy and social influence increased intention, with social influence stronger; functional and endowment-based threats increased resistance, and assistive configurations were preferred over replacement-oriented ones [7].
Not shown in excerpt	2026	Emergency Medicine Journal	10.1136/emermed-2025-215220	CoDE-HF machine-learning decision-support tool	Emergency department clinicians across three acute care hospitals [8]	UTAUT	Performance expectancy, effort expectancy, social influences, facilitating conditions [8]	Clinicians saw the tool as useful for objective communication and reprioritising missed diagnoses; output presentation was important for acceptance, while limited computer access and processing time were the main barriers [8].
Not shown in excerpt	2026	International Journal of Africa Nursing Sciences	10.1016/j.ijans.2026.101147	AI-based CDSS for nursing practice	320 registered nurses in Egyptian university hospitals [9]	Adapted UTAUT questionnaire	Psychological safety, professional identity, acceptance of AI-based CDSS [9]	Psychological safety was the strongest direct predictor of acceptance; professional identity was correlated with acceptance but did not mediate the relationship once psychological safety was controlled [9].

Bottom line

Across these studies, **acceptance is driven less by the label "AI" or "CDSS" itself than by whether clinicians perceive the tool as trustworthy, workflow-compatible, transparent, and decision-supportive rather than replacement-oriented** [2] [3] [6] [7] [8]. The most common measured constructs are **perceived usefulness / performance expectancy, perceived ease of use / effort expectancy, trust, social influence, and workflow fit**, and the most consistent finding is that these factors shape willingness to use the tool more than demographic characteristics alone [1] [3] [7] [8] [9].

Confidence level: **High** — multiple direct empirical studies across different clinical settings and years converge on the same acceptance pattern, although the retrieved excerpts do not expose author names.

References

1. Chien, S.-C., Chien, C.-H., Chen, C.-Y., (...), Li, Y.-C. (2025) Using technology acceptance model to explore physicians' perspectives of clinical decision support system alerts. *BMJ Health and Care Informatics*. https://doi.org/10.1136/bmjhci-2024-101128
2. Chien, Shuo-Chen, Chien, Chia-Hui, Chen, Chun-You, (...), Li, Yu-Chuan (2025) Using technology acceptance model to explore physicians' perspectives of clinical decision support system alerts. *BMJ Health and Care Informatics*. https://doi.org/10.1136/bmjhci-2024-101128

3. Herter, W.E., Khuc, J., Bonten, T.N., (...), Chavannes, N.H. (2026) Usability and Usefulness of Machine Learning–Based Clinical Decision Support Software in Primary Care: Survey of Users in a Prospective Observational Study. *JMIR Medical Informatics*. https://doi.org/10.2196/80527
4. Berge, G. T., Granmo, O. C., Tveit, T. O., (...), Sharma, J. (2024) Machine learning-driven clinical decision support system for concept-based searching: a field trial in a Norwegian hospital. *BMC Medical Informatics and Decision Making*. https://doi.org/10.1186/s12911-023-02101-x
5. Ji, M., Genchev, G.Z., Huang, H., (...), Yu, G. (2021) Evaluation framework for successful artificial intelligence–enabled clinical decision support systems: Mixed methods study. *Journal of Medical Internet Research*. https://doi.org/10.2196/25929
6. Stalp, Jan Lennart, Schneider, Juliane Alexandra, Krause, Anna, (...), Wolff, Dominik (2026) Physicians' acceptance of large language model–based clinical decision support tools in gynecologic oncology: a technology acceptance model study. *Frontiers in Digital Health*. https://doi.org/10.3389/fdgth.2026.1896184
7. Zhu, J., Wu, X., Zhang, C., (...), Bian, X. (2026) AI-CDSS adoption among healthcare professionals: the influence of endowment-based autonomy threats and configuration preferences. *Health Policy and Technology*. https://doi.org/10.1016/j.hlpt.2026.101286
8. https://www.sciencedirect.com/science/article/pii/S1472021326001042
9. Abousoliman, A.D., Atta, M.H.R., Hamed, H.M. (2026) Acceptance of AI-based clinical decision support systems among registered nurses: psychological safety and the mediating role of professional identity in a cross-sectional study. *International Journal of Africa Nursing Sciences*. https://doi.org/10.1016/j.ijans.2026.101147